ASSIGNMENT REPORT

Assignment 4

JavaScript Interactive Website

Five connected front-end features for the Apex Motors site: form validation, FAQ accordion, contact pop-up, page background control, and a live date/time display.

TEAM

Apex Motors

GROUP

SE-2507

MEMBERS

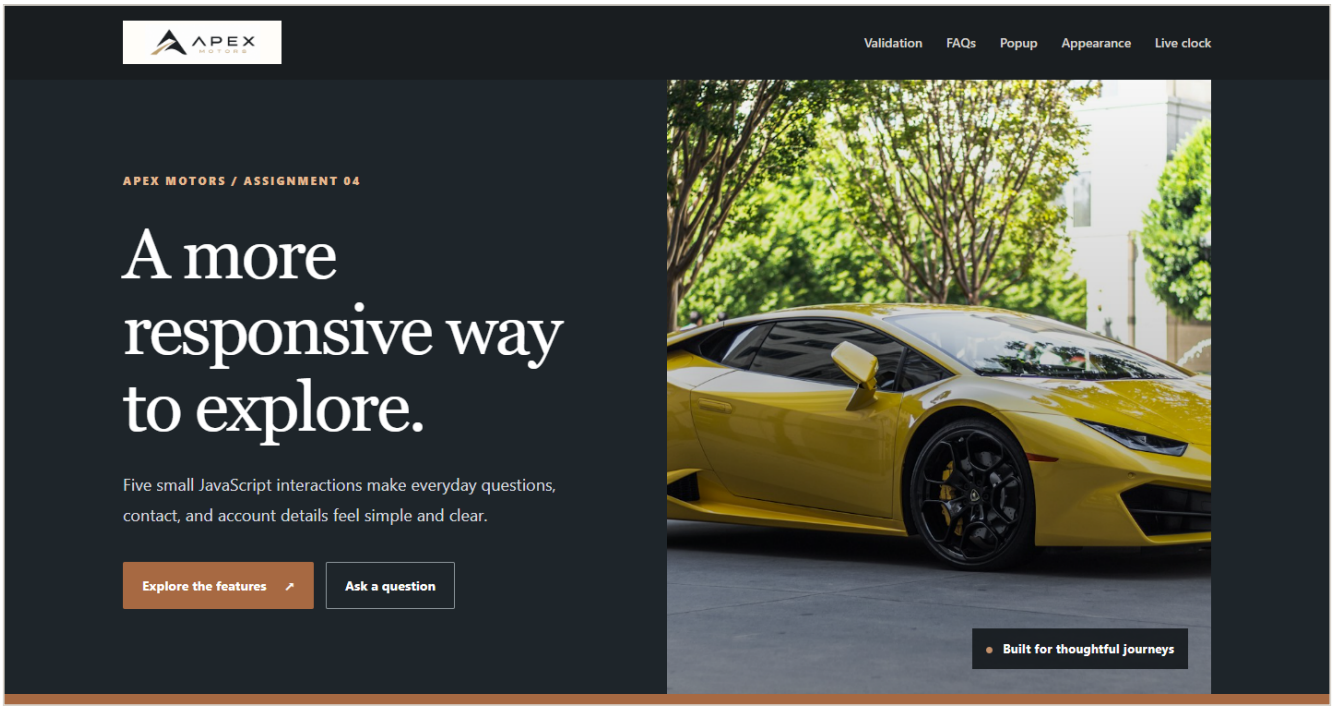
Arsen, Kuttibay, Farabi, Muhammad

COMPONENT LEADS FOR THE PRACTICE DEFENSE

Kuttibay	Account form validation	form-validation.js
Farabi	FAQ accordion	faq.js
Arsen	Contact pop-up	popup.js
Muhammad	Color cycle and live clock	experience.js

DEPLOYED WEBSITE

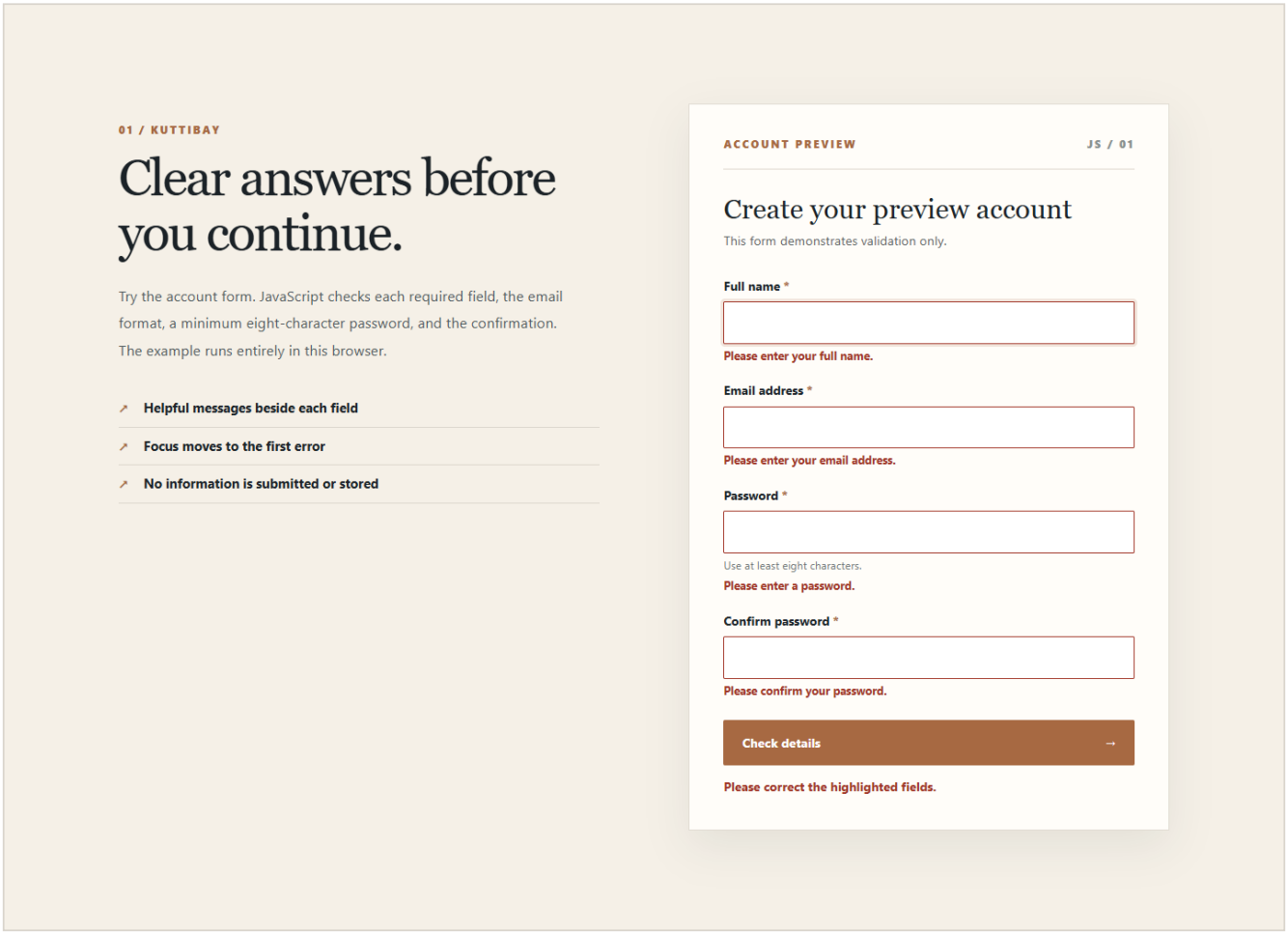
https://vertex-blip.github.io/Car-dealership-website/assignment4-javascript/



TASK 01 / KUTTIBAY

Form validation

The account preview checks every required value and gives a specific message next to each invalid field. The first invalid field receives focus so the visitor can fix it immediately.



IMPLEMENTATION STEPS

01 Create labeled full name, email, password, and confirmation controls in **index.html**; link **js/form-validation.js** with **defer**.

02 Stop default form submission and check blank values, an email pattern, a password of at least eight characters, and an exact confirmation match.

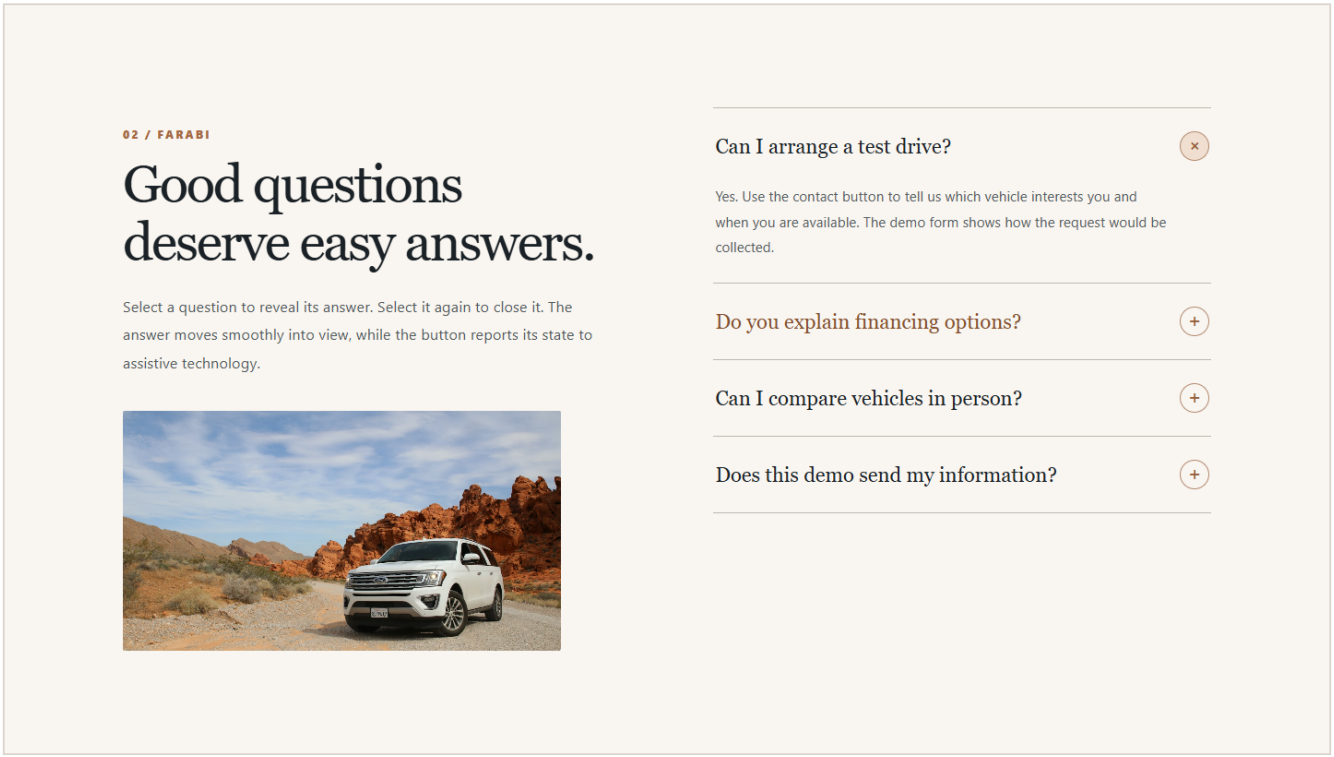
03 Write field-specific feedback, set **aria-invalid**, and focus the first invalid input. Clear messages as the user corrects values.

04 Show a local success message when all values pass; no account is created and no data is sent.

TASK 02 / FARABI

FAQ accordion

Four dealership questions begin collapsed. A click reveals the selected answer; another click hides it. The transition is smooth while reduced-motion preferences are respected.



IMPLEMENTATION STEPS

01 Mark each question as a real **button** and pair it with an answer using **aria-controls** and a unique ID.

02 Start each answer collapsed in CSS and set its accessible hidden state in **js/faq.js**.

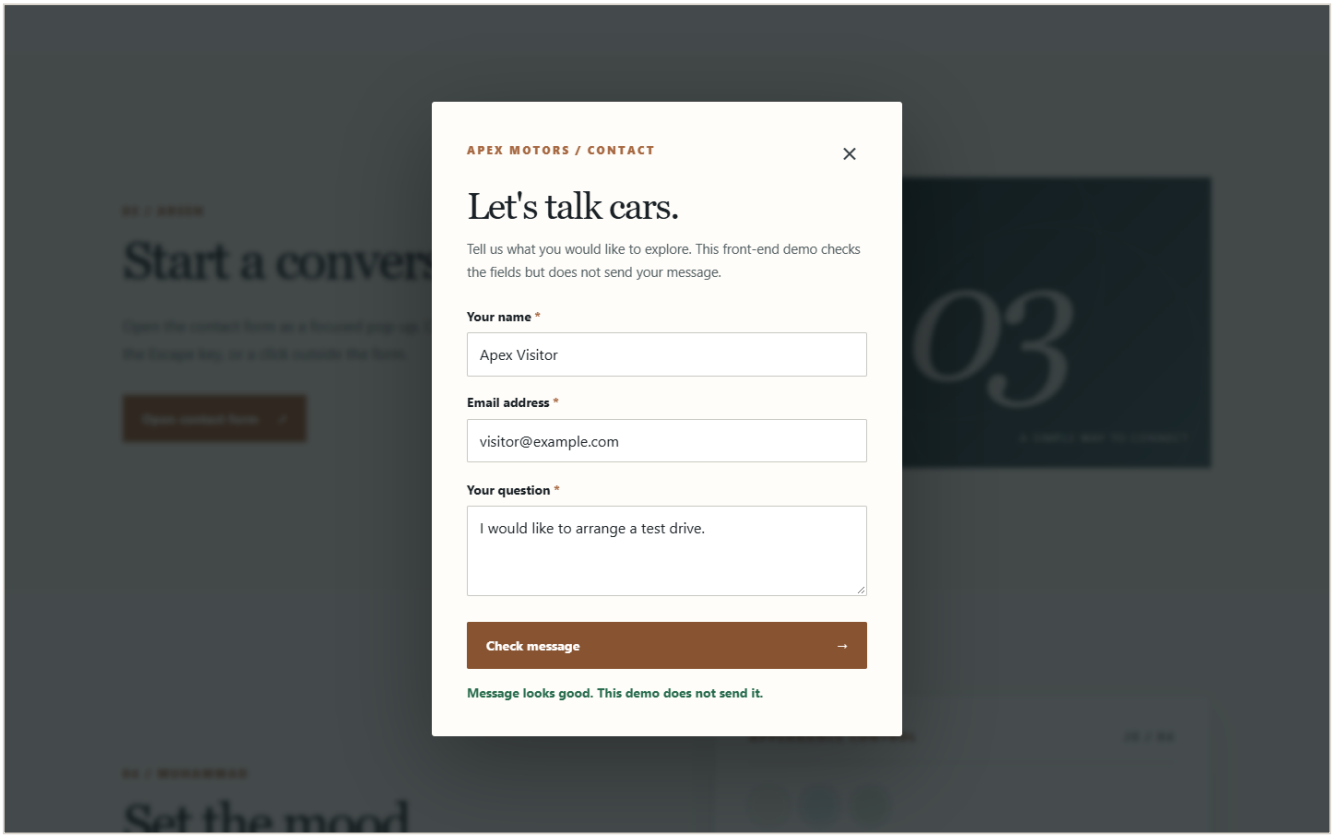
03 On click, toggle **aria-expanded**, **aria-hidden**, **inert**, and the **is-open** class.

04 Animate **grid-template-rows** between **0fr** and **1fr**; a second click reverses the same behavior.

TASK 03 / ARSEN

Contact pop-up

The contact form stays hidden until a button opens it. Visitors can close it with the close button, Escape, or a click outside the pop-up.



IMPLEMENTATION STEPS

01 Place the labeled contact form inside a native **<dialog>** element and keep it closed initially.

02 Call **showModal()** from either contact button and focus the first form field.

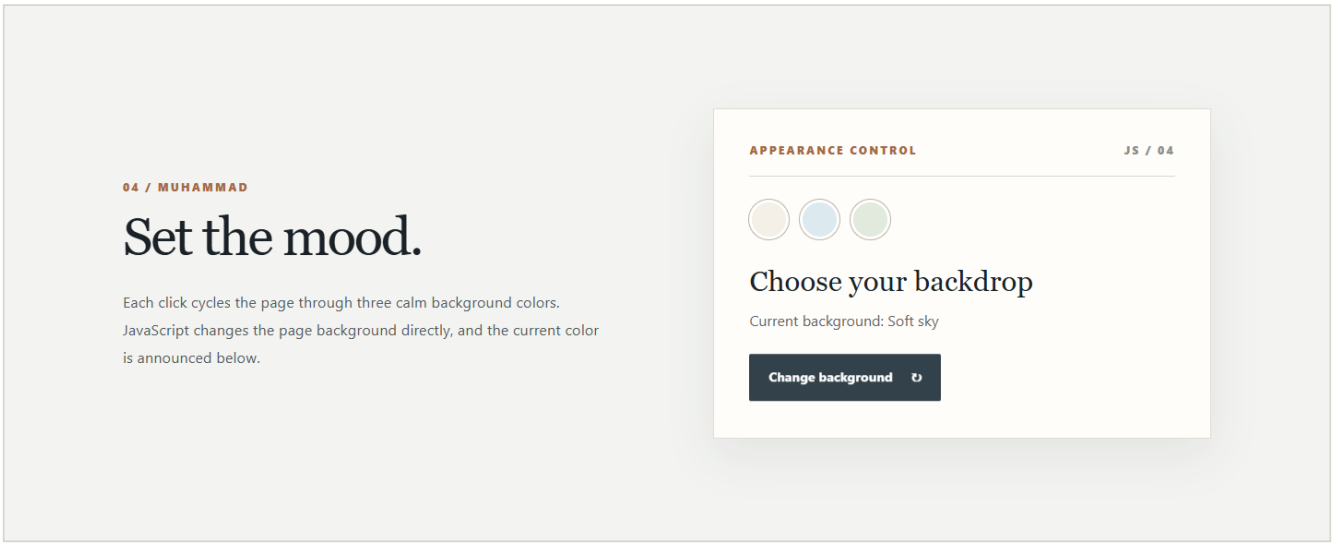
03 Call **close()** from the close button or when a click lands outside the dialog rectangle. The browser handles Escape.

04 Validate required name, email, and message fields locally; restore focus to the opener when the dialog closes.

TASK 04 / MUHAMMAD

Change background color

A button cycles the page background through warm cream, soft sky, and pale sage. The active color name updates beside the control.



01 Store the three colors in an array and advance an index on each button click.

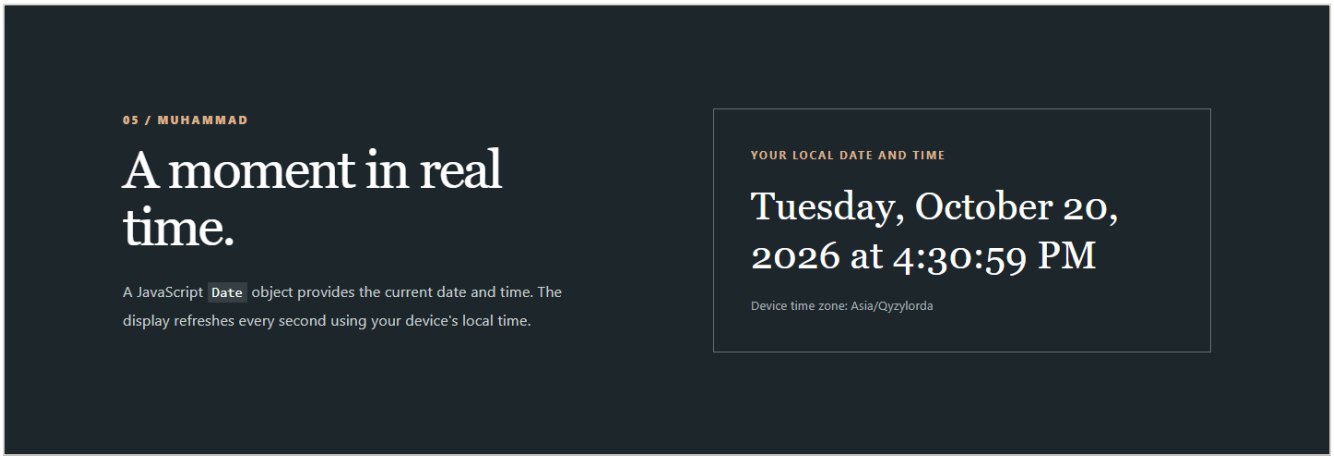
02 Assign the selected value to **document.body.style.backgroundColor**; CSS animates the change.

03 Update the nearby status text so the current selection is clear.

TASK 05 / MUHAMMAD

Display current date and time

The clock shows the viewer's local date and time in a readable format and refreshes every second.



01 Create a fresh **Date**, format it with **Intl.DateTimeFormat**, and place it in a semantic **<time>** element.

02 Call the update immediately and every 1,000 ms with **setInterval**; show the device time zone.

VERIFICATION

Tested as a complete page

We checked the actual browser behavior and the linked files, including the narrow-screen layout. The forms are deliberately front-end demonstrations and never send or store information.

AREA	CHECK	RESULT
Form	Empty fields, invalid email, short password, mismatch, valid input	**Pass**
FAQ	Initially closed; click open; click closed; state attributes	**Pass**
Pop-up	Open; close button; outside click; Escape; local validation	**Pass**
Background	Three changes return to the initial color	**Pass**
Clock	Readable local time; timestamp changes after one second	**Pass**
Layout	Chrome and Edge desktop; Chrome mobile at 390 px; no overflow	**Pass**

WHAT WE LEARNED

JavaScript adds behavior to the HTML and CSS structure from the earlier assignments. Validation needs precise feedback, an accordion needs a clear open state, and a pop-up needs several reliable ways to close. Small features work better when their state is visible to both sighted users and assistive technology.

The four script files keep responsibilities clear for the practice defense. Every member can explain their assigned component and should also rehearse the others. The site uses only local assets, so the same folder can run from the ZIP or GitHub Pages.

SUBMISSION

Each of the four members must upload this report and the project ZIP through the LMS and defend the work at the practice lesson. Project URL: https://vertex-blip.github.io/Car-dealership-website/assignment4-javascript/